

HARMAN SINGH MALHOTRA

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SUMMARY

Software Engineer with 1.5+ years building and operating backend microservices and full-stack products in TypeScript, Node.js, and React. Took two greenfield services from empty repository to production ownership on Google Cloud Platform (GCP), across distributed systems, RESTful Application Programming Interface (API) design.

EDUCATION

Bachelor of Technology, Computer Science (AI/ML) | UPES, Dehradun | 8.78 CGPA Aug 2021 - Jun 2025
Indian School Certificate (ISC), Class XII | St. Joseph's Academy, Dehradun | 92% Jun 2021

EXPERIENCE

Software Engineer, promoted from Software Engineer Intern Dec 2024 - Present
Loop Health (Invoq Health) Bengaluru, IN

- Architected Loop's first communications microservice, dispatching **1M+ emails and WhatsApp messages under a minute at peak** via 1,000 record batches with back-pressure across SendGrid and Sinch on Cloud Tasks.
- Instrumented traceability over **100% of outbound messages**, capturing opens, clicks, and SMTP failures, replacing a fire-and-forget API with a campaign/job/message/event model and priority state machine.
- Designed a premium Rater microservice on a JSON rules engine with set operators and pro-rata calculators, scaling pricing inputs **from 3 attributes to 18** and retiring hardcoded per-policy functions.
- Consolidated two diverging copies of member validation into one three-phase Validation Service serving **40,000 to 50,000 records/day**, onboarding the legacy pipeline via an adapter behind feature flags.
- Surfaced silent member drop-offs previously caught only through customer escalations, streaming Cloud Pub/Sub events into BigQuery **across 3 services** and automating insurer endorsement generation.

Web Developer Intern Jun 2024 - Aug 2024
Digital Jalebi Dehradun, IN

- Built **Venti**, an Event Tech platform, from scratch on Next.js and Node.js, spanning event management, user onboarding, communications, and booth analytics, with the MVP running across **12+ exhibits**.
- Cut exhibit load times by **1,600ms** with an **on-premise Express and MongoDB server** that served asset bundles over HTTPS, streamed session state via WebSockets, and ran without venue internet.

Project Intern, Part-time Jun 2024 - Jul 2024
HERE Technologies Dehradun, IN

- Engineered a Python and Scrapy crawler spanning **12+** live sources, extracting location records that fed directly into HERE's mapping data pipeline for positional accuracy checks.
- Raised data refresh frequency by replacing a manual collection process with a scheduled framework that normalised and deduplicated records across all **12+** tracked sources.

PROJECTS

U-Tour, Autonomous campus tour booking platform [Live](#) | [GitHub](#)

- Implemented the scheduling gateway for a LiDAR guided tour robot, using **Redis-backed Bull queues** to serialise bookings, merge overlapping tours, and block double-booking (Node.js, Express).
- Modelled tours, sessions, and users over Prisma/PostgreSQL behind Firebase authentication, triggering SendGrid status notifications from templated queue workers on booking and cancellation events.

GraphPathGuru, Graph algorithm visualiser [Live](#) | [GitHub](#)

- Ported **6** shortest-path algorithms (Dijkstra, Bellman-Ford, Floyd-Warshall, Johnson, SPFA, Yen) from C++ subprocesses into one **Rust WebAssembly module**, unblocking deployment of the MERN visualiser.
- Animated step-by-step execution on a ReactFlow editor with per-user graph persistence over MongoDB.

SKILLS

Programming Languages: TypeScript, JavaScript, Python, C++, SQL, HTML/CSS

Frameworks: Node.js, Express.js, FastAPI, React.js, Next.js, Prisma ORM, REST APIs, Microservices, Event-Driven Architecture, Redux, Zustand, TailwindCSS

Databases & Cloud: PostgreSQL, Redis, MongoDB, Google BigQuery, Firestore, Supabase, Google Cloud Platform (GCP), Cloud Run, Cloud Tasks, Cloud Pub/Sub, Secret Manager

Tools & Practices: Git, Docker, GitHub Actions, Jira, Jest, Swagger/OpenAPI, Feature Flags (Flagsmith), Agile/Scrum, Performance Optimization, Unit & Integration Testing, System Design, Data Structures & Algorithms

CERTIFICATIONS AND ACHIEVEMENTS

Azure AI Engineer Associate, Microsoft | **LeetCode**, 700+ problems at a **1570** rating in C++

Graph Theory Contest Winner, UPES | **Associate Design & VFX Head**, UPES CSI, led **9** designers over **20+** projects